5 Selected Exercises[1]

The following exercises are grouped into sections according to the chapters of *Exercises in Introductory Physics.* In parentheses the location of the corresponding subject matter in *The Feynman Lectures on Physics,* Volumes I–III, is provided. For example, the subject matter of the exercises in Section 5-1, "Conservation of energy, statics (Vol. I, Ch. 4)" is discussed in *The Feynman Lectures on Physics*, Volume I, Chapter 4.

Within each section the exercises are subdivided into categories according to degree of difficulty. In the order in which they appear in each section, these are: easy exercises (*), intermediate exercises (**), and more sophisticated and elaborate exercises (***). The average student should have little trouble solving the easy exercises, and should be able to solve most of the intermediate exercises within a reasonable time—perhaps ten to twenty minutes each. The more sophisticated exercises generally require a deeper physical insight or more extensive thought, and will be of interest principally to the better student.

5-1 **Conservation of energy, statics (Vol. I, Ch. 4)**

*** 1-1** A ball of radius 3.0 cm and weight 1.00 kg rests on a plane tilted at an angle α with the horizontal and also touches a vertical wall. Both surfaces have negligible friction. Find the force with which the ball presses on each plane.

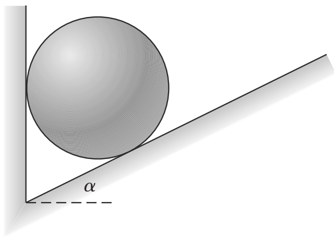

FIGURE 1-1